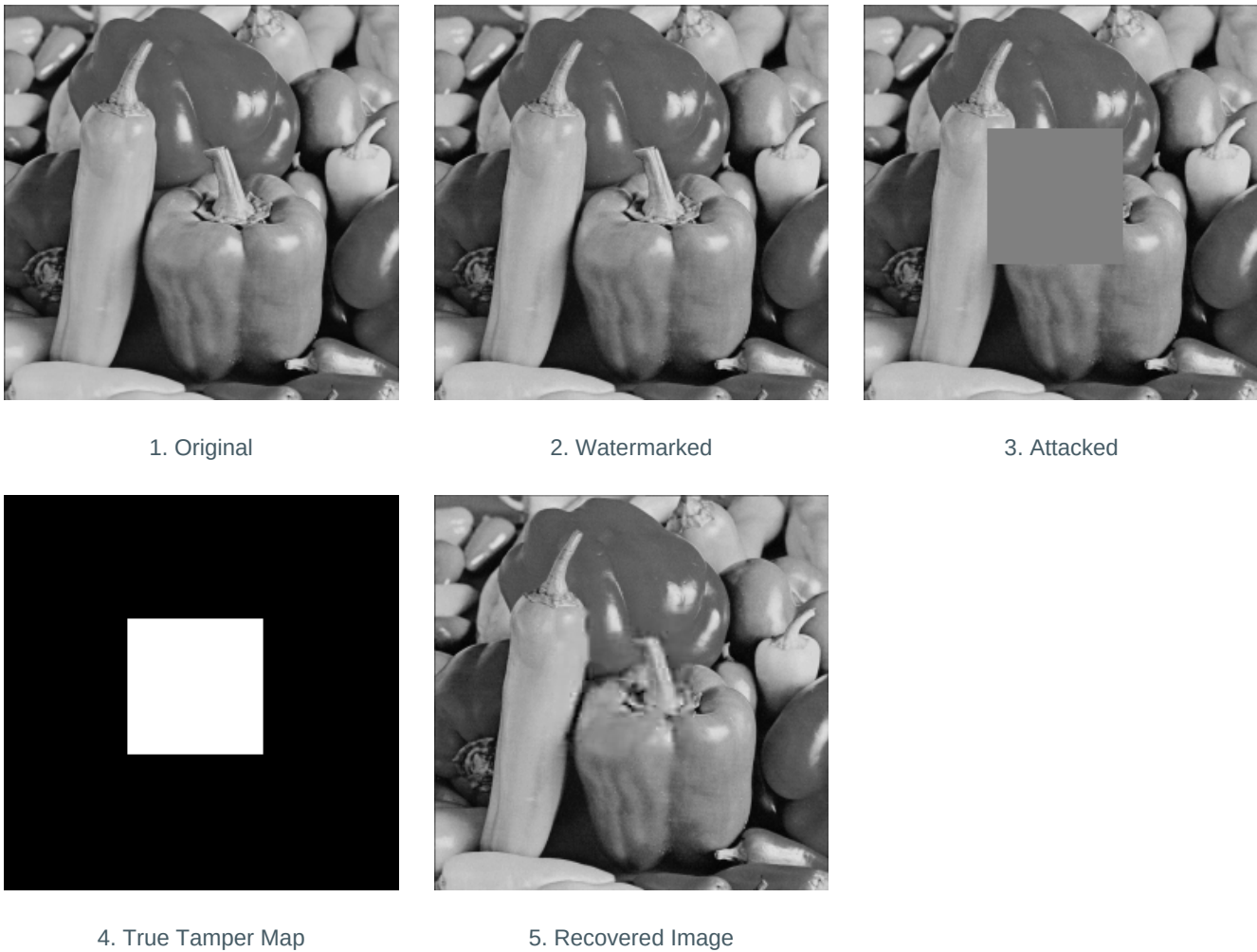


Fragile Watermarking: Visual Evaluation Report

Generated: May 04, 2026 21:02 | Total Experiments: 32

Experiment 1: Peppers — Attack: Crop 33%



Metric	Value	Metric	Value
Watermarked PSNR	44.16 dB	Recovered PSNR	34.48 dB
Watermarked SSIM	0.9861	Recovered SSIM	0.9688
True Positive Rate (TPR)	1.0000	Recovered NC	0.9959
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 2: Peppers — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.16 dB	Recovered PSNR	38.05 dB
Watermarked SSIM	0.9861	Recovered SSIM	0.9795
True Positive Rate (TPR)	0.5000	Recovered NC	0.9982
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 3: Peppers — Attack: Salt_and_Pepper_1%



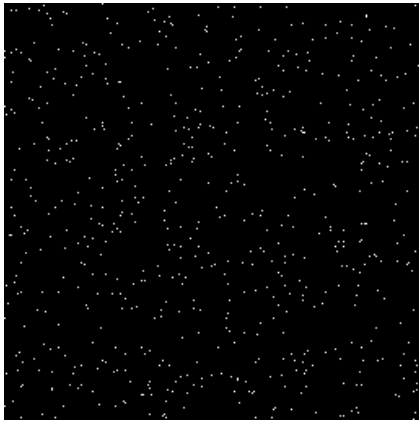
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.16 dB	Recovered PSNR	25.88 dB
Watermarked SSIM	0.9861	Recovered SSIM	0.8557
True Positive Rate (TPR)	0.5773	Recovered NC	0.9704
False Positive Rate (FPR)	0.0430	Tamper F1-Score	0.1932

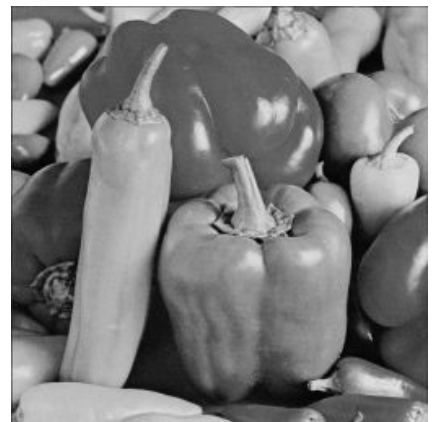
Experiment 4: Peppers — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

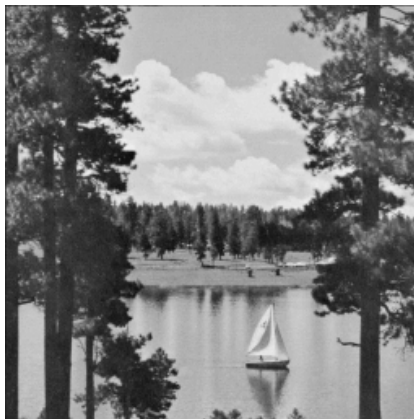
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.16 dB	Recovered PSNR	37.31 dB
Watermarked SSIM	0.9861	Recovered SSIM	0.9622
True Positive Rate (TPR)	N/A	Recovered NC	0.9979
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

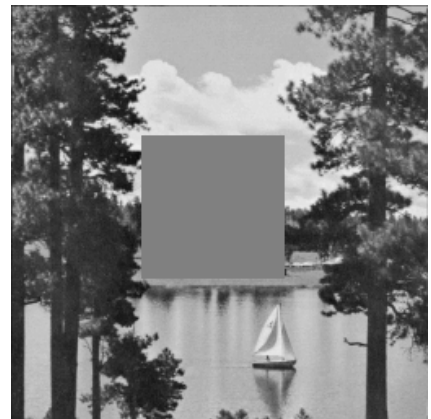
Experiment 5: Sailboat on lake — Attack: Crop 33%



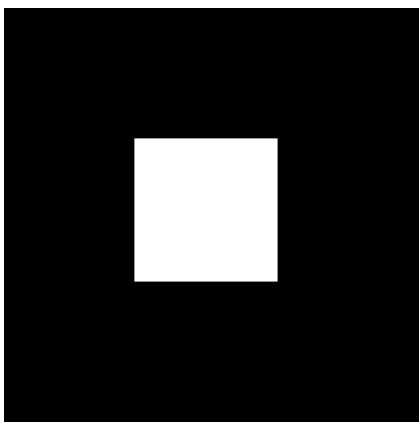
1. Original



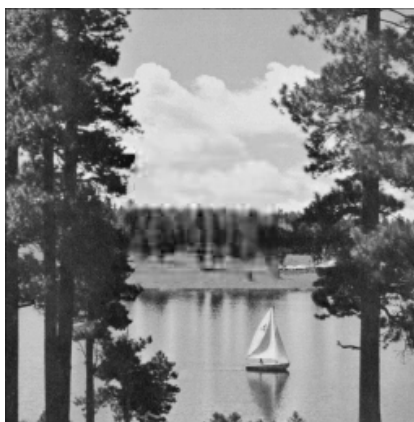
2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.16 dB	Recovered PSNR	34.76 dB
Watermarked SSIM	0.9885	Recovered SSIM	0.9709
True Positive Rate (TPR)	1.0000	Recovered NC	0.9974
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 6: Sailboat on lake — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



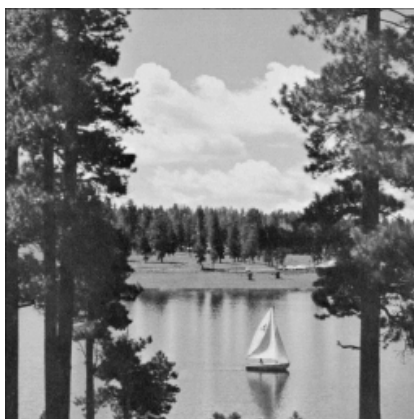
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.16 dB	Recovered PSNR	38.83 dB
Watermarked SSIM	0.9885	Recovered SSIM	0.9810
True Positive Rate (TPR)	0.5000	Recovered NC	0.9990
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

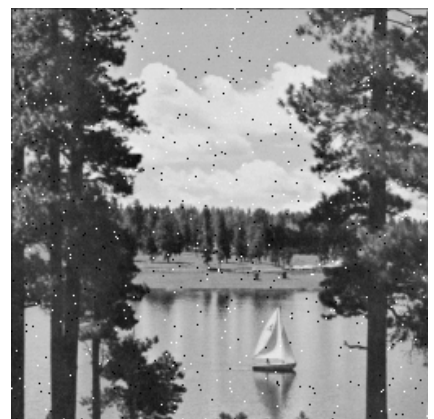
Experiment 7: Sailboat on lake — Attack: Salt_and_Pepper_1%



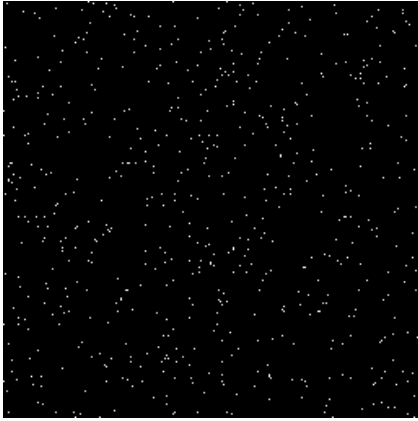
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.16 dB	Recovered PSNR	25.72 dB
Watermarked SSIM	0.9885	Recovered SSIM	0.8699
True Positive Rate (TPR)	0.5735	Recovered NC	0.9791
False Positive Rate (FPR)	0.0416	Tamper F1-Score	0.1932

Experiment 8: Sailboat on lake — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked

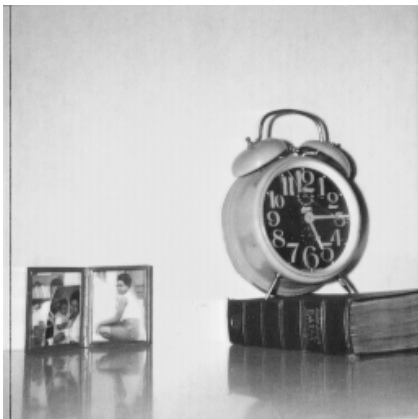


4. True Tamper Map

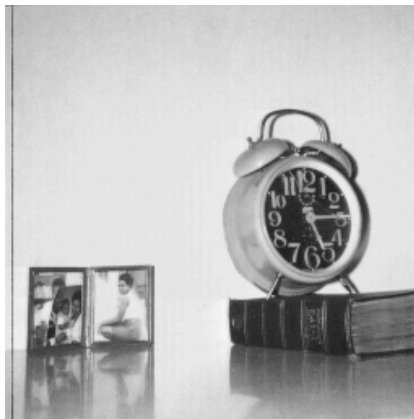
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.16 dB	Recovered PSNR	34.57 dB
Watermarked SSIM	0.9885	Recovered SSIM	0.9561
True Positive Rate (TPR)	N/A	Recovered NC	0.9973
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

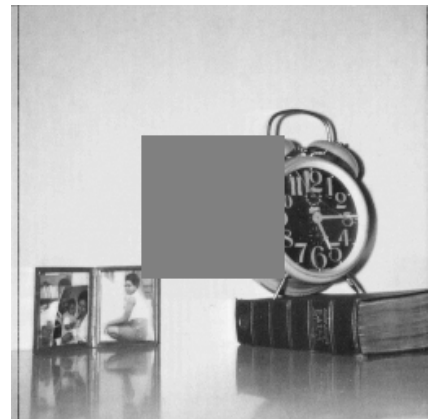
Experiment 9: Clock — Attack: Crop 33%



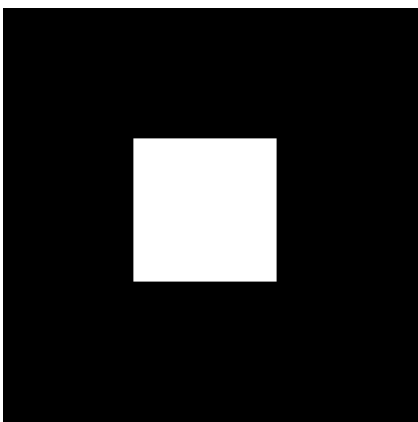
1. Original



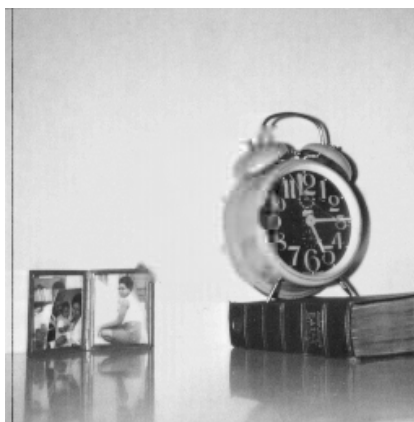
2. Watermarked



3. Attacked



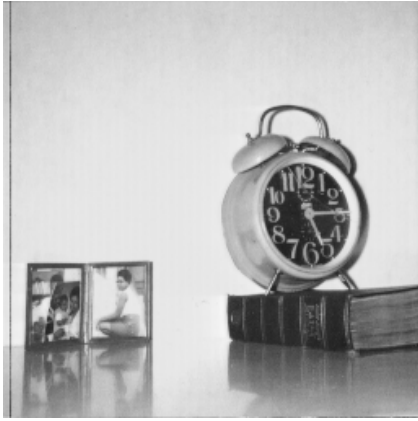
4. True Tamper Map



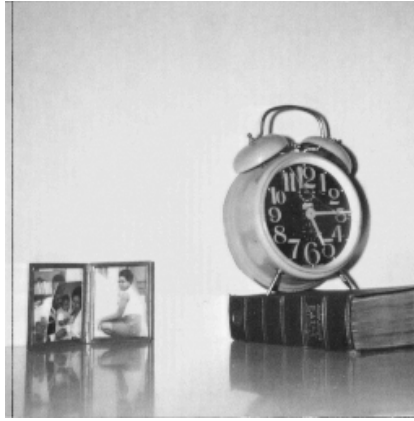
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.22 dB	Recovered PSNR	34.19 dB
Watermarked SSIM	0.9779	Recovered SSIM	0.9682
True Positive Rate (TPR)	1.0000	Recovered NC	0.9962
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 10: Clock — Attack: Copy-Move



1. Original



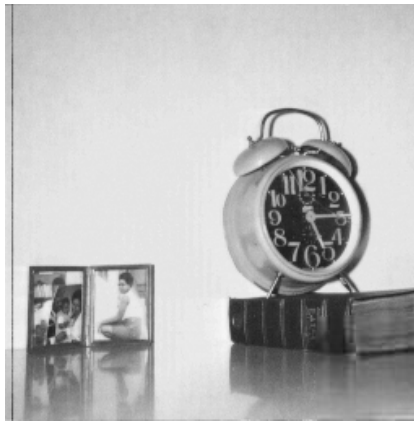
2. Watermarked



3. Attacked



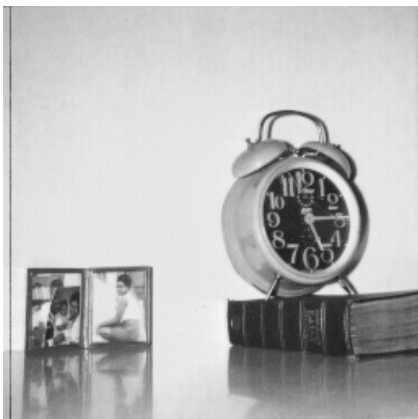
4. True Tamper Map



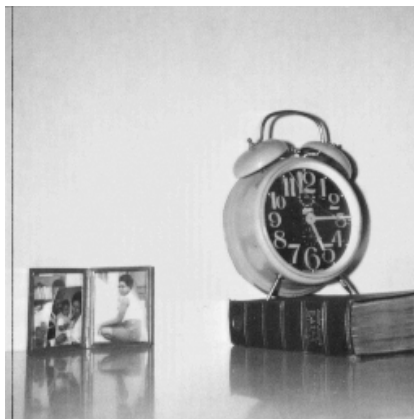
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.22 dB	Recovered PSNR	41.11 dB
Watermarked SSIM	0.9779	Recovered SSIM	0.9712
True Positive Rate (TPR)	0.5000	Recovered NC	0.9992
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 11: Clock — Attack: Salt_and_Pepper_1%



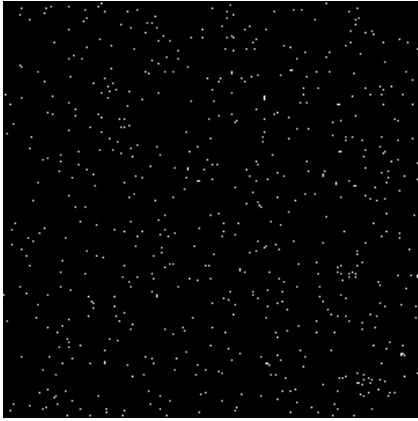
1. Original



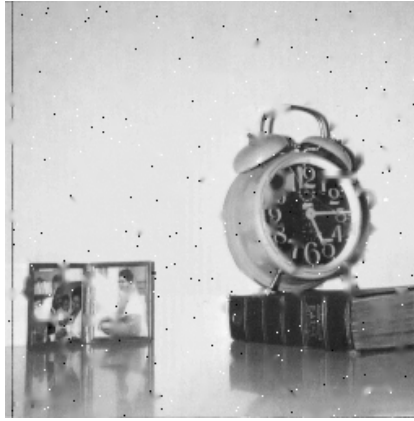
2. Watermarked



3. Attacked



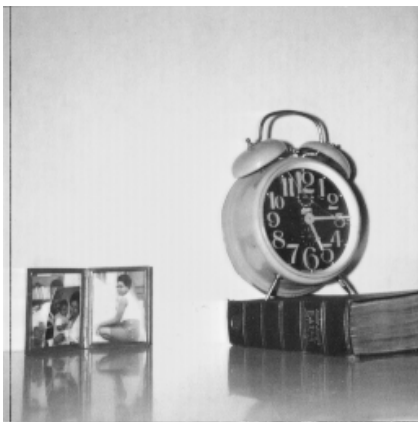
4. True Tamper Map



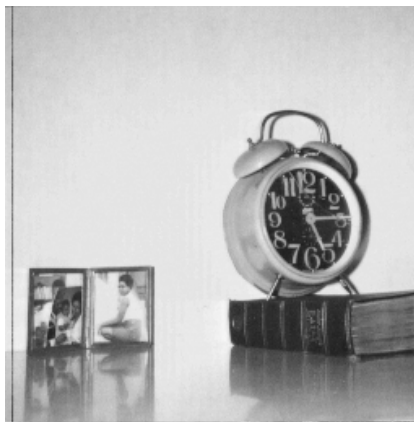
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.22 dB	Recovered PSNR	25.52 dB
Watermarked SSIM	0.9779	Recovered SSIM	0.8624
True Positive Rate (TPR)	0.5680	Recovered NC	0.9720
False Positive Rate (FPR)	0.0421	Tamper F1-Score	0.1929

Experiment 12: Clock — Attack: JPEG_80



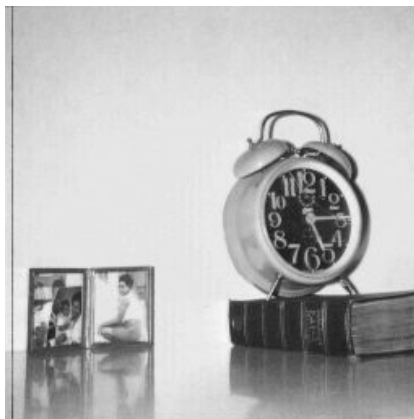
1. Original



2. Watermarked



3. Attacked

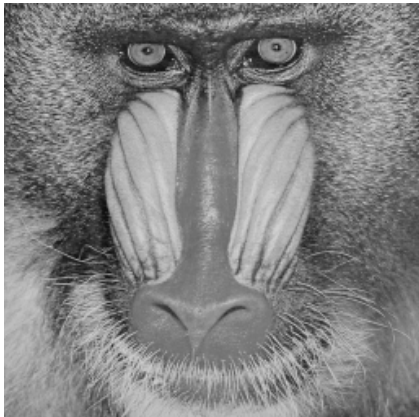


4. True Tamper Map

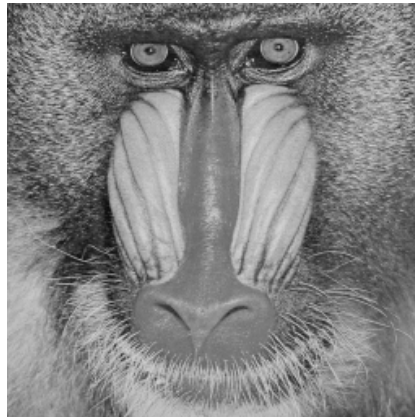
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.22 dB	Recovered PSNR	38.07 dB
Watermarked SSIM	0.9779	Recovered SSIM	0.9664
True Positive Rate (TPR)	N/A	Recovered NC	0.9985
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

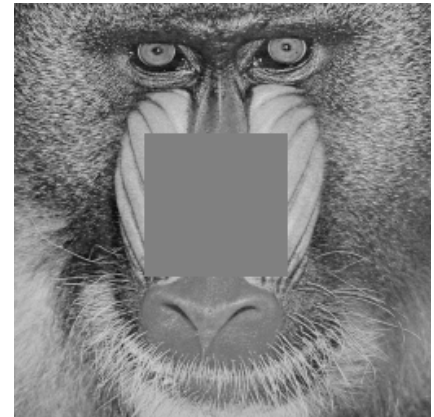
Experiment 13: Mandrill — Attack: Crop 33%



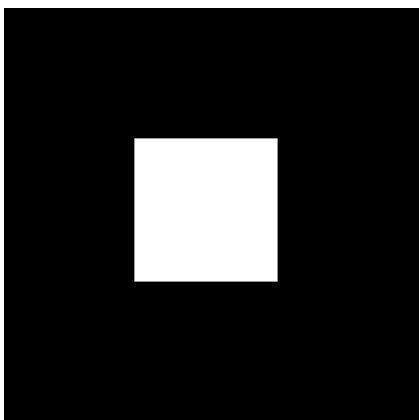
1. Original



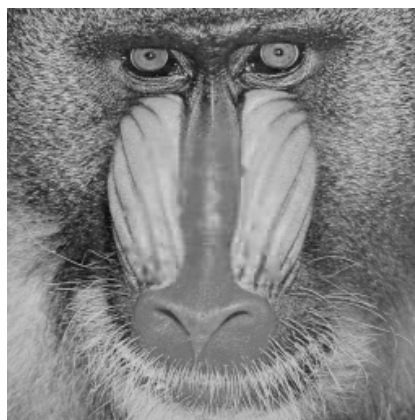
2. Watermarked



3. Attacked



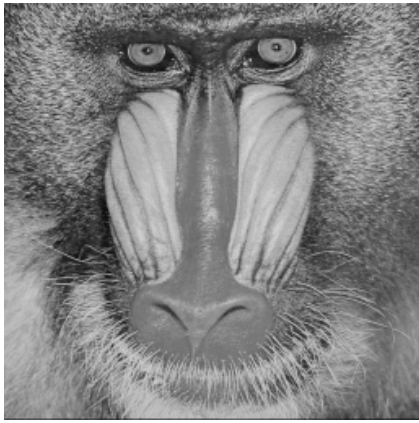
4. True Tamper Map



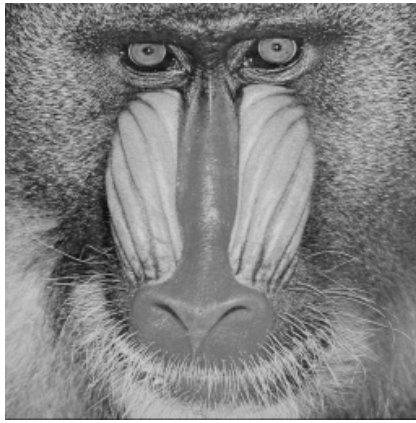
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	36.40 dB
Watermarked SSIM	0.9945	Recovered SSIM	0.9701
True Positive Rate (TPR)	1.0000	Recovered NC	0.9950
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

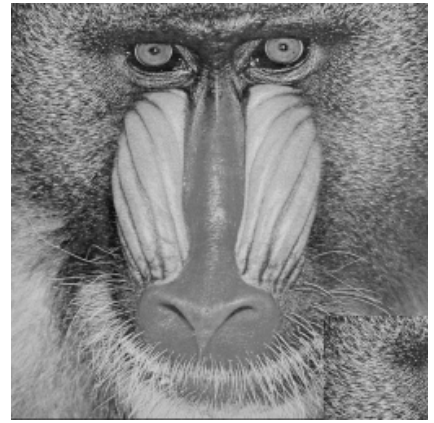
Experiment 14: Mandrill — Attack: Copy-Move



1. Original



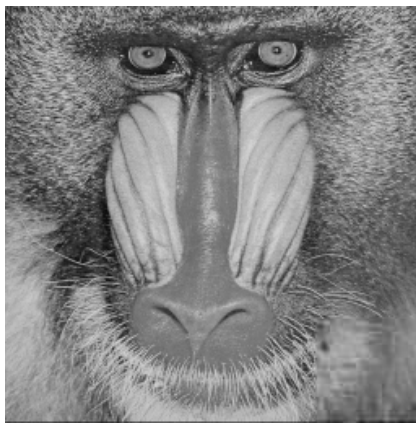
2. Watermarked



3. Attacked



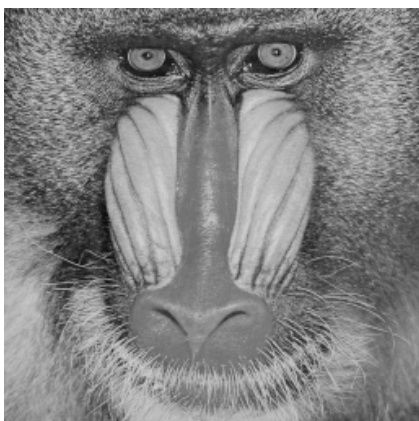
4. True Tamper Map



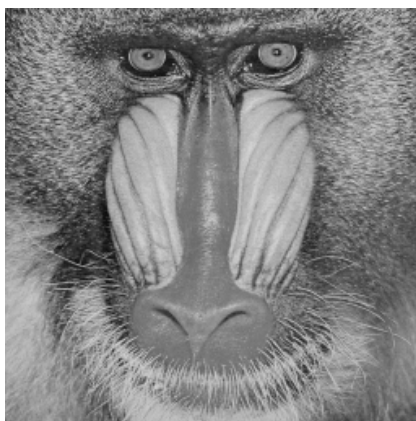
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	37.70 dB
Watermarked SSIM	0.9945	Recovered SSIM	0.9781
True Positive Rate (TPR)	0.5000	Recovered NC	0.9963
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

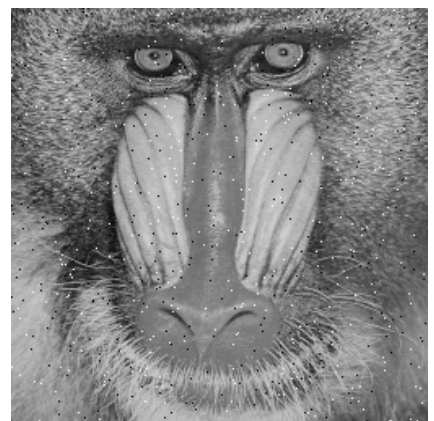
Experiment 15: Mandrill — Attack: Salt_and_Pepper_1%



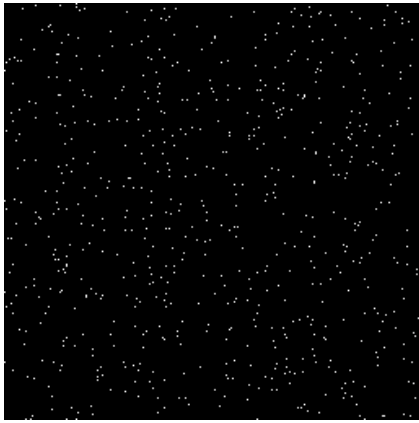
1. Original



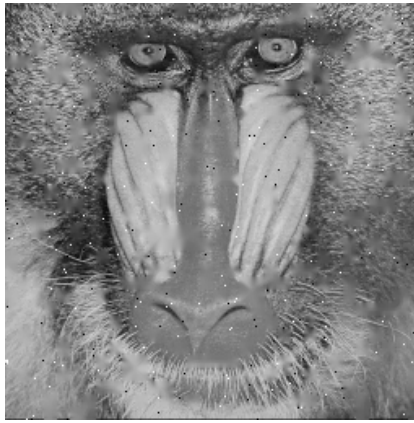
2. Watermarked



3. Attacked



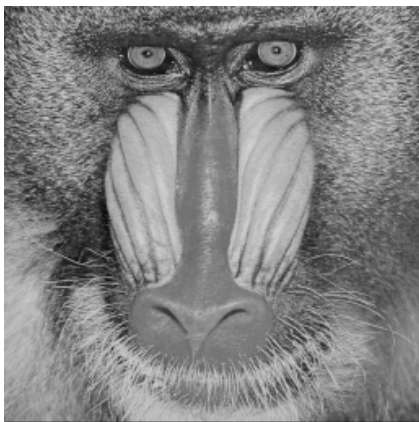
4. True Tamper Map



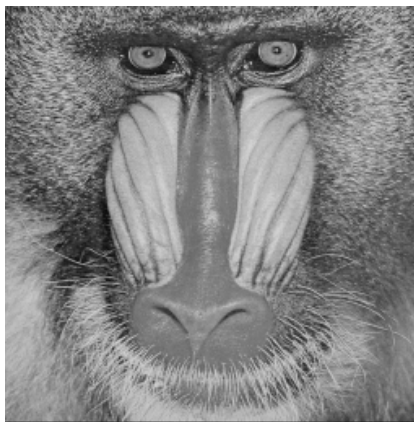
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	25.75 dB
Watermarked SSIM	0.9945	Recovered SSIM	0.8251
True Positive Rate (TPR)	0.5786	Recovered NC	0.9415
False Positive Rate (FPR)	0.0464	Tamper F1-Score	0.1915

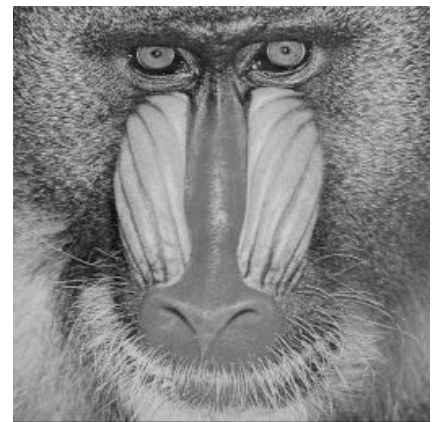
Experiment 16: Mandrill — Attack: JPEG_80



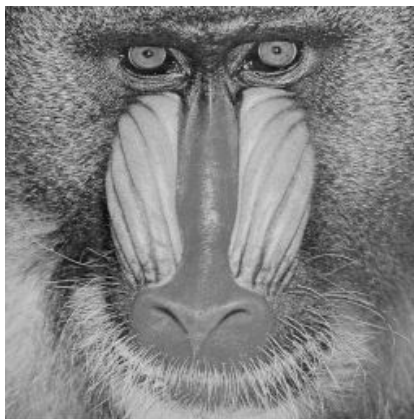
1. Original



2. Watermarked



3. Attacked



5. Recovered Image

4. True Tamper Map

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	32.12 dB
Watermarked SSIM	0.9945	Recovered SSIM	0.9437
True Positive Rate (TPR)	N/A	Recovered NC	0.9868
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

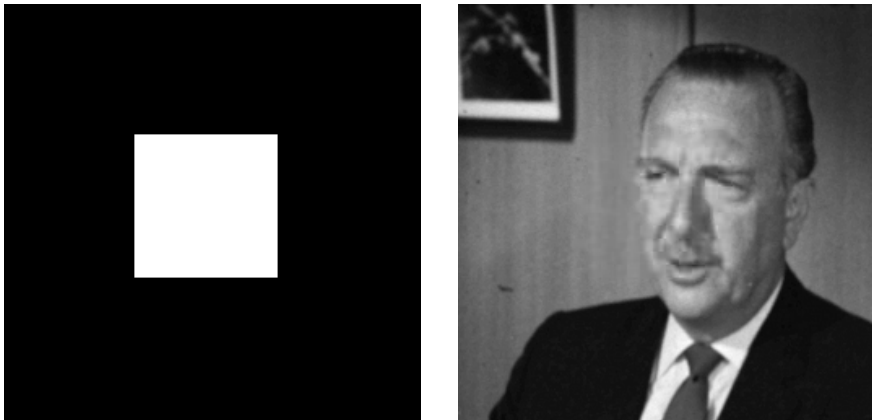
Experiment 17: Walter Cronkite — Attack: Crop 33%



1. Original

2. Watermarked

3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.06 dB	Recovered PSNR	38.88 dB
Watermarked SSIM	0.9773	Recovered SSIM	0.9680
True Positive Rate (TPR)	1.0000	Recovered NC	0.9986
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 18: Walter Cronkite — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.06 dB	Recovered PSNR	44.02 dB
Watermarked SSIM	0.9773	Recovered SSIM	0.9784
True Positive Rate (TPR)	0.5000	Recovered NC	0.9996
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 19: Walter Cronkite — Attack: Salt_and_Pepper_1%



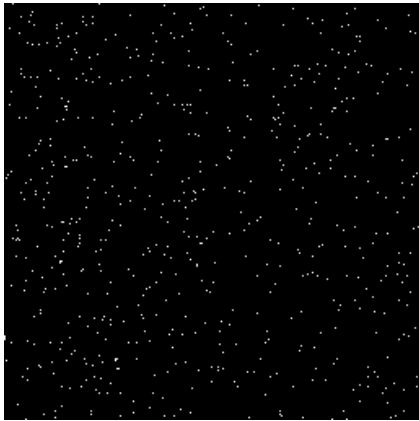
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.06 dB	Recovered PSNR	27.95 dB
Watermarked SSIM	0.9773	Recovered SSIM	0.8630
True Positive Rate (TPR)	0.6140	Recovered NC	0.9824
False Positive Rate (FPR)	0.0458	Tamper F1-Score	0.1932

Experiment 20: Walter Cronkite — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.06 dB	Recovered PSNR	42.31 dB
Watermarked SSIM	0.9773	Recovered SSIM	0.9735
True Positive Rate (TPR)	N/A	Recovered NC	0.9994
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

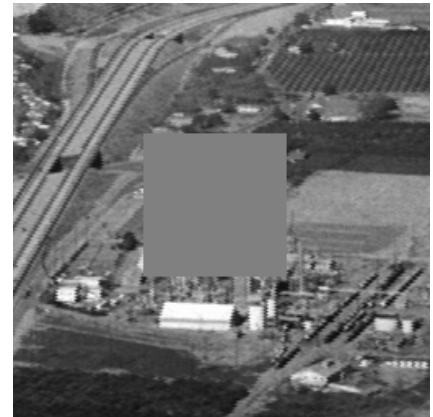
Experiment 21: Chemical plant — Attack: Crop 33%



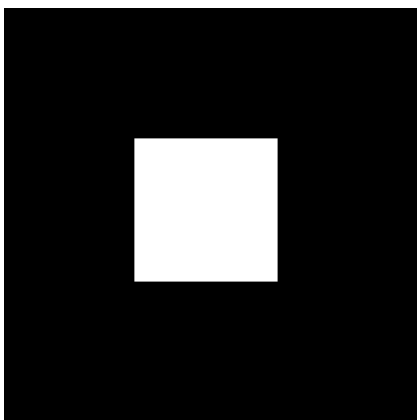
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	33.56 dB
Watermarked SSIM	0.9929	Recovered SSIM	0.9657
True Positive Rate (TPR)	1.0000	Recovered NC	0.9920
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 22: Chemical plant — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	36.14 dB
Watermarked SSIM	0.9929	Recovered SSIM	0.9826
True Positive Rate (TPR)	0.5000	Recovered NC	0.9956
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 23: Chemical plant — Attack: Salt_and_Pepper_1%



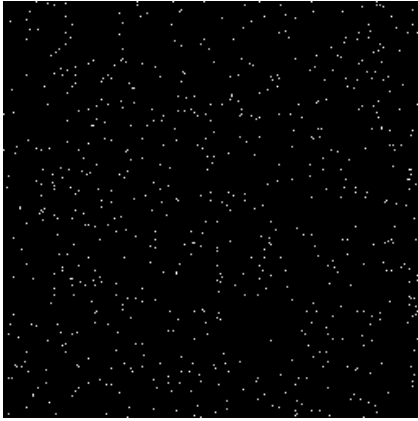
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	25.91 dB
Watermarked SSIM	0.9929	Recovered SSIM	0.8551
True Positive Rate (TPR)	0.5667	Recovered NC	0.9536
False Positive Rate (FPR)	0.0441	Tamper F1-Score	0.1936

Experiment 24: Chemical plant — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.11 dB	Recovered PSNR	35.15 dB
Watermarked SSIM	0.9929	Recovered SSIM	0.9628
True Positive Rate (TPR)	N/A	Recovered NC	0.9945
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

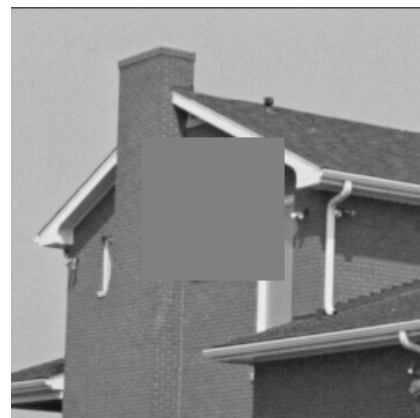
Experiment 25: House — Attack: Crop 33%



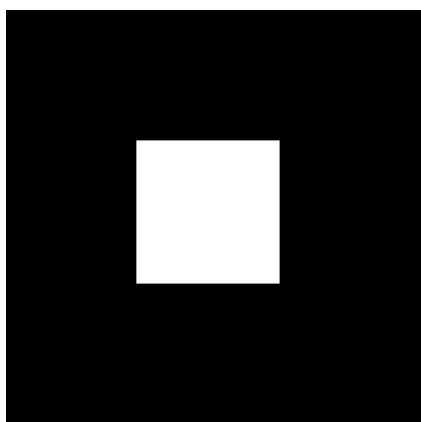
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.12 dB	Recovered PSNR	35.09 dB
Watermarked SSIM	0.9791	Recovered SSIM	0.9511
True Positive Rate (TPR)	1.0000	Recovered NC	0.9953
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 26: House — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



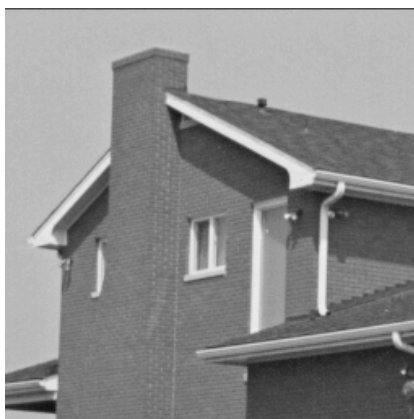
5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.12 dB	Recovered PSNR	39.00 dB
Watermarked SSIM	0.9791	Recovered SSIM	0.9711
True Positive Rate (TPR)	0.5000	Recovered NC	0.9981
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 27: House — Attack: Salt_and_Pepper_1%



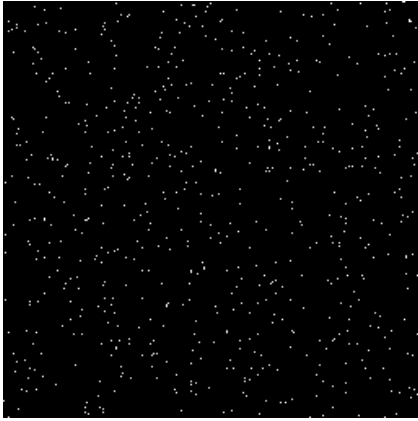
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.12 dB	Recovered PSNR	26.90 dB
Watermarked SSIM	0.9791	Recovered SSIM	0.8273
True Positive Rate (TPR)	0.5852	Recovered NC	0.9688
False Positive Rate (FPR)	0.0463	Tamper F1-Score	0.1939

Experiment 28: House — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.12 dB	Recovered PSNR	38.49 dB
Watermarked SSIM	0.9791	Recovered SSIM	0.9470
True Positive Rate (TPR)	N/A	Recovered NC	0.9978
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A

Experiment 29: Airplane — Attack: Crop 33%



1. Original

2. Watermarked

3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.18 dB	Recovered PSNR	31.84 dB
Watermarked SSIM	0.9827	Recovered SSIM	0.9593
True Positive Rate (TPR)	1.0000	Recovered NC	0.9897
False Positive Rate (FPR)	0.0055	Tamper F1-Score	0.9799

Experiment 30: Airplane — Attack: Copy-Move



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.18 dB	Recovered PSNR	36.46 dB
Watermarked SSIM	0.9827	Recovered SSIM	0.9731
True Positive Rate (TPR)	0.5000	Recovered NC	0.9964
False Positive Rate (FPR)	0.0021	Tamper F1-Score	0.6602

Experiment 31: Airplane — Attack: Salt_and_Pepper_1%



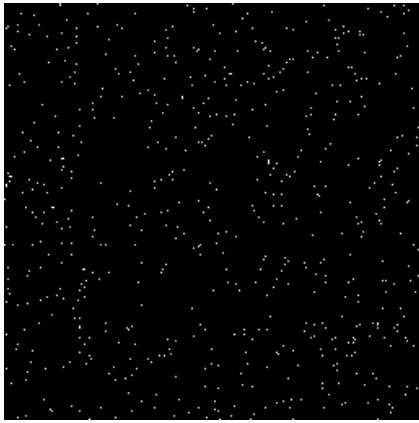
1. Original



2. Watermarked



3. Attacked



4. True Tamper Map



5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.18 dB	Recovered PSNR	26.20 dB
Watermarked SSIM	0.9827	Recovered SSIM	0.8605
True Positive Rate (TPR)	0.5514	Recovered NC	0.9623
False Positive Rate (FPR)	0.0404	Tamper F1-Score	0.1911

Experiment 32: Airplane — Attack: JPEG_80



1. Original



2. Watermarked



3. Attacked



4. True Tamper Map

5. Recovered Image

Metric	Value	Metric	Value
Watermarked PSNR	44.18 dB	Recovered PSNR	36.38 dB
Watermarked SSIM	0.9827	Recovered SSIM	0.9629
True Positive Rate (TPR)	N/A	Recovered NC	0.9964
False Positive Rate (FPR)	N/A	Tamper F1-Score	N/A
